SOLUTIONS

CHEMISTRY :

1. (D)

CH_3 and aniline are planar

2. (A) [Concept Code : C110507/504]

$$H_3AsO_4 + OH^- \longrightarrow H_2AsO_4^- + H_2O$$

Initial milimols	30	10	0
After reaction	20	0	10

$$H_3AsO_4 + HAsO_4^{2-} \longrightarrow 2H_2AsO_4^-$$

Initial milimols	20	30	10
After reaction	0	10	50

So final pH = $pKa_2 + \log\frac{10}{50} = 6.3$

3. (C)

A=> propyne B=> mesitylele C => acetone D => mesitylene E=> phorone

4. (C)

Kp ($CaCO_3$) = 4 atm

Kp ($BaCO_3$) = 6 atm

Pressure above ($CaCO_3$+$BaCO_3$) = 6 atm

Total pressure = 6+1+2= 9 atm

5. (ABC)

	N_2 +	PCl_5 $\rightleftharpoons$	PCl_3 +	Cl_2
Moles at t = 0	1	3	0	0
Moles at equilibrium	1	(3 – x)	x	x

Total moles present at equilibrium = 4 + x

Given total pressure at equilibrium = 2.05

Total volume = 100 Litre

Θ PV = nRT

$\therefore$ 2.05 × 100 = (4 + x) × 0.082 × 500

$\therefore$ x = 1.0

Degree of dissociation 'α' for PCl_5

$= \frac{\text{Moles dissociated}}{\text{Total moles present}} = \frac{1}{3} = 0.3333 = 33.33\%$

Also $K_P = \frac{x^2}{(3-x)}\left[\frac{P}{\sum n}\right]^2 = \frac{x^2}{(3-x)} \times \frac{2.05}{(4+x)}$

$= \frac{(1)^2 \times 2.05}{(3-1)(4+1)}$

K_P= 0.205 atm.

6. (CD)

7. (BC)

At equilibrium

$$\underset{0.1}{H_2S} \overset{K_{a_1}=10^{-5}}{\rightleftharpoons} \underset{0.1}{HS^-} + \underset{10^{-5}}{H^+}$$

$$\underset{0.1}{HS^-} \overset{K_{a_2}=10^{-8}}{\rightleftharpoons} \underset{10^{-4}}{S^{2-}} + \underset{10^{-5}}{H^+}$$

$$NiS_{(S)} \overset{K_{sp}=10^{-20}}{\rightleftharpoons} \underset{10^{-4}}{S^{2-}} + \underset{10^{-16}}{Ni^{2+}}$$

8. (ACD)

9. (ABC)

Compounds formed

Total = 10

Optically active =10

Optically inactive=0

Diastereomer pairs = 4

Enantiomer pairsm= 5

10. (D) [Concept Code : C110402]

When solids $A_{(s)}$, $E_{(s)}$ and $P_{(s)}$ are kept together then

$$A_{(s)} \overset{Kp_1=4atm^2}{\rightleftharpoons} \underset{p_1}{C_{(g)}} + \underset{p_1+p_2+p_3}{B_{(g)}}$$

$$E_{(s)} \overset{Kp_2=32atm^2}{\rightleftharpoons} \underset{p_2}{D_{(g)}} + \underset{p_1+p_2+p_3}{B_{(g)}}$$

$$P_{(s)} \overset{Kp_3=5atm}{\rightleftharpoons} Q_{(s)} + \underset{p_1+p_2+p_3}{B_{(g)}}$$

$p_1 = \frac{4}{5}$, $p_2 = \frac{32}{5}$, $p_3 = -1$

But $p_3 = -1$ means to establish $P_{(s)} \rightleftharpoons B_{(g)} + Q_{(s)}$, it has to go backward which is not possible due to absence of $Q_{(s)}$ in the container so $p_3 = 0$

Hence $(p_1 + p_2)p_1 = 4, (p_1 + p_2)p_2 = 32$

$\Rightarrow p_1 + p_2$ = Partial pressure of $B_{(g)}$ = 6 atm

and the total pressure = 12 atm

(Integer Type)

1. (9)

In order to prevent precipitation of ZnS,

$[Zn^{2+}]\,[S^{2-}] < K_{sp}(ZnS) = 1 \times 10^{-21}$

(Ionic product)

or $(0.1).[S^{2-}] < 1 \times 10^{-21}$

or $[S^{2-}] < 1 \times 10^{-20}$

This is the maximum value of $[S^{2-}]$ before ZnS will precipitate. Let $[H^+]$ to maintain this $[S^{2-}]$ be x.

Thus for $H_2S \rightleftharpoons 2H^+ + S^{2-}$,

$$K_a = \frac{[H^+]^2[S^{2-}]}{[H_2S]} = \frac{x^2(1\times10^{-20})}{0.1}$$

$= 1.0 \times 10^{-21}$

or $x = [H^+] = 0.1$ M. => pH= 1

when $[S^{2-}] = 1 \times 10^{-20}$ M and $[Cd^{2+}][S^{2-}] = K_{sp}(CdS) = 8 \times 10^{-27}$

Then $[Cd^{2+}] = 8 \times 10^{-7}$ M

So, x=1 and y=8 => x+y = 9

$\therefore$ No ZnS will precipitate at a concentration of H^+ greater than 0.1 M.

2. (4)

$x^2+y^2+2z^2 = 2yz + 2x + 4z - 5$

=> x = 1; z = y = 2

3. (9)

3 + 4 + 2 = 9

4. 8

$PCl_3 + Cl_2 \rightleftharpoons PCl_5$

0.20 0.08 0.40 moles/litre

Kc = 25

If 0.22 mole of Cl_2 is added then at equilibrium

0.20 – x 0.30 – x 0.40 + x

$$25 = \frac{0.40+x}{(0.20-x)(0.30-x)}$$

or x = 0.1

$[PCl_3] = 0.2 - 0.1 = 0.1$ moles

$[Cl_2] = 0.3 - 0.1 = 0.2$ moles

$[PCl_5] = 0.4 + 0.1 = 0.5$ moles

Sum of conc. = 0.8

5. (8)

P is $[BH_2(NH_3)_2]^+[BH_4]^-$ Q is borazine

6. (5)

MATHS

1. $y^2 = 4ax$. The normal chord at (4a, 4a) subtends a right angle at focus.

2. Tangents at $(a\cos\alpha, b\sin\alpha)$ and $(a\cos\beta, b\sin\beta)$

meet at $\left(\dfrac{a\cos\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}, \dfrac{b\sin\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha-\beta}{2}\right)}\right) \equiv (h,k)$

$\dfrac{k}{h} = \dfrac{b}{a}\tan\left(\dfrac{\alpha+\beta}{2}\right) = \text{constant} \Rightarrow y = \left[\dfrac{b}{a}\tan\left(\dfrac{\alpha+\beta}{2}\right)\right]x$

3. Asymptotes of a rectangular hyperbola are perpendicular

$(x - y - 1) + \lambda(3x - 4y - 6) = 0 \Rightarrow \text{slope} = \dfrac{-4}{3}$

$4x + 3y = 17$

4. $y = mx$ in $ax^2 + 2hxy + by^2 = 0$

$\Rightarrow x^2[a + (2h)m + bm^2] = 0$

$m_1 = km_2 \qquad m_1 + m_2 = \dfrac{-2h}{b} \qquad m_1m_2 = \dfrac{a}{b}$

5. $y = mx + \sqrt{a^2m^2 - b^2}$

$\Rightarrow \sqrt{9p^2 - 8} = 1 \Rightarrow p^2 = 1$

6. $t = y^2$, First order linear differential equation in variable t.

7. The length of perpendicular from centre of the circle on both the lines will be equal.

8. Length of perpendicular from centre on the line will be greater than the radius of the circle

9. $\int_0^{\alpha}(x - x^3)\,dx = \dfrac{1}{2}\int_0^{1}(x - x^3)\,dx$

10. First order linear D.E. in $\dfrac{dx}{dy}$

Integer :

1. $\dfrac{dy}{dx} - \left(\dfrac{2x}{1+x^2}\right)y = (x^2 - 1)$

2. The line joining origin and (2,3) will be parallel to the axis of the parabola

4. Sum of focal distances for any point = 8

$A : (2 + 4\cos(\theta), \sqrt{12}\sin(\theta))$

5. $\dfrac{(x^2 + y^2 + 1)}{2} + (x - y) = k$

$C_1C_2 = r_1 + r_2$

6. $hx - ky = h^2 - k^2$ becomes a tangent to $y^2 = 4ax$

$\Rightarrow y = \left(\dfrac{h}{k}\right)x + \left(\dfrac{k^2 - h^2}{k}\right)$

$$\Rightarrow \frac{h(k^2-h^2)}{k^2} = a$$

SOLUTIONS (PHYSICS)

1) (B)

$$P = \frac{\varepsilon^2 R}{(r+R)^2} = \frac{\varepsilon^2 n}{(r+nR)^2} R$$

$$\Rightarrow \quad r = R\sqrt{n}$$

2) (B)

The current density J, electric field E at the cross-section are related by

$$J = \frac{E}{\rho}$$

multiplying both side by A.

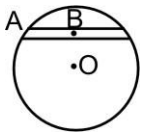
$$JA = \frac{EA}{\rho} \text{ or } I = \frac{\phi}{\rho} \text{ Where } \phi \text{ is electric flux.}$$

$$JA = \frac{EA}{\rho} \text{ or } I = \frac{\phi}{\rho} \quad \text{or} \quad \phi = \rho I$$

3) (A)

In the position of balance no current exists through the galvanometer, and hence through the cell. Therefore the magnitude of r does not affect the condition for balance

4. (b)



If point B is being crossed then the particle will reach to other end

$$\Delta KE = (V_B - V_A)q$$

$$V_B = K\rho\pi R^2 \times \frac{11}{6}$$

$$V_A = K\rho\pi R^2 \times \frac{4}{3}$$

5. BCD

With the help of loop law & junction law

$$V_a - V_b = +16V$$

Hence b, c, d are correct

6. AD

7. ACD

8. AB

w.r.t. trolley, $\vec{F} = 0$ & $\vec{\tau}_{cm} = 0$

9. AC

$$-\frac{q}{4} - 4 - \frac{q}{2} + 8 = 0$$

$$\Rightarrow q = \frac{16}{3}$$

at $t = \infty \quad i = 0$

at $t = 0 \Rightarrow -4 - 2i + 8 = 0 \Rightarrow i = 2A$

10. ABC

MATRIX MATCHING :

1. A–t; B–t; C–p; D–p
From theory

2. (A) P,Q,R, S, T; (B) P,Q,R,S,T; (C) P,Q, S,T; (D) P,Q,R,S,T

3. a-S, b-P, c-T, d-R

$V - ig(G+R)$

$= (100)(10^{-6})(100 + 2 \times 10^5)$

$\simeq 20V$

$\frac{i - i_g}{i_g} = \frac{G}{S}$

$i = i_g + \left(\frac{G}{S}\right) i_g$

$= 100(10^{-6}) + \left(\frac{100}{1}\right)(100)(10^{-6})$

$= 0.1 \times 10^{-3} + 10(10^{-3})$

$\simeq 10mA$.

22. **A-Q, B-S, C-P, D-R**

$\Delta q_E = \varepsilon_1 \left(C'_{eq} - C_q\right)$. $C^1_{eq} \rightarrow$ equation capacitance with S_1 closed

$= 12\left[3 - \frac{8}{3}\right] = 4\mu C$

$\Delta q_S = 12\mu C$

$\Delta q_E = \varepsilon_1 \left(C^4_{eq} - C_{eq}\right),\ 12\left[4 - \frac{4}{3}\right] = 32\ \mu C$

$\Delta q_S = 12 \times 4 = 48\ \mu C$

NUMERICAL :

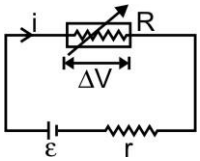
1. (1)

The electron covers a distance $\frac{d}{2}$ across the first capacitor when if comes out of it.

$\therefore$ in second capacitor it has to cover $\frac{d}{2}$ across it.

$\therefore V_1 = V_2$

2. 4
3. **2**
4. **5**



When i = 0 and $\Delta V = 10V$

$\varepsilon = 10V$ ...(i)

When $\Delta V = 0, R = 0$ and i = 2A

$\therefore \quad \varepsilon = ir$

$r = \frac{10}{2} = 5$

5. (2)

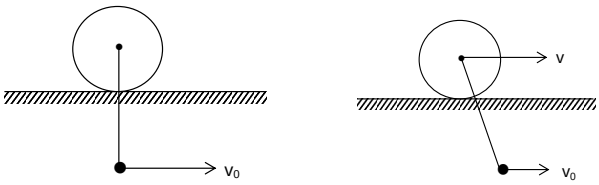
E inside shaded portion is

ℓ = distance between centres

$$E = \frac{\rho\ell}{3\varepsilon_0} = \frac{\varepsilon_0 \times 6 - 10 + 10^{-2}}{3 \times \varepsilon_0} = 2 \times 10^{-2} N/C$$

$$\Rightarrow \frac{E_0}{100} = \frac{2}{100} \Rightarrow E_0 = 2$$

6. 3



$mV_0 = (M + m) V$

$$V = \frac{mV_0}{M+m}$$

From energy conservation

$$\frac{1}{2} mV_0^2 + Mg\ell = mgh + Mg\ell + \frac{1}{2} (M+m)V^2$$

$$\frac{1}{2} mV_0^2 - \frac{1}{2} \frac{m^2 v_0^2}{(M+m)} = mgh$$

$$\frac{1}{2} \frac{mV_0^2[M+m-m]}{[M+m]} = mgh$$

$$h = \frac{1}{2} \frac{V_0^2}{g} \frac{M}{(M+m)}$$